

Motivation

Industries, companies, cities, households, etc., all consume energy. Whether opting for electricity, gas, or thermal power – or, more likely, a combination of them – the need for energy is all around us. Both consumers and producers can benefit greatly from accurate estimates of future consumption, not in the least because extreme volatility of wholesale prices forces market parties to hedge against volume risk and price risk. Handling on incorrect volume estimates is often expensive. The ability to accurately forecast future energy consumption is a determining factor of the financial performance of market players. Therefore, the forecasts are also a key input of the decision-making process.

Problem Statement

Your task is to build a machine learning pipeline for energy consumption where you will:

- Plot and analyze the data and check the data quality
- Build a model and forecast for the next N hours
- Identify the normal and anomalous behavior on the time-series dataset that could act as early indicators for quality degradation.
- Provide evaluation metrics for you model(s).

Details

- Dataset from: Jan 2013 – Dec 2014
- Frequency: Hourly
- Target: Load (Mwh)
- Predictors: Load, Temperature, Cloudiness, Irradiation, Public Holidays